

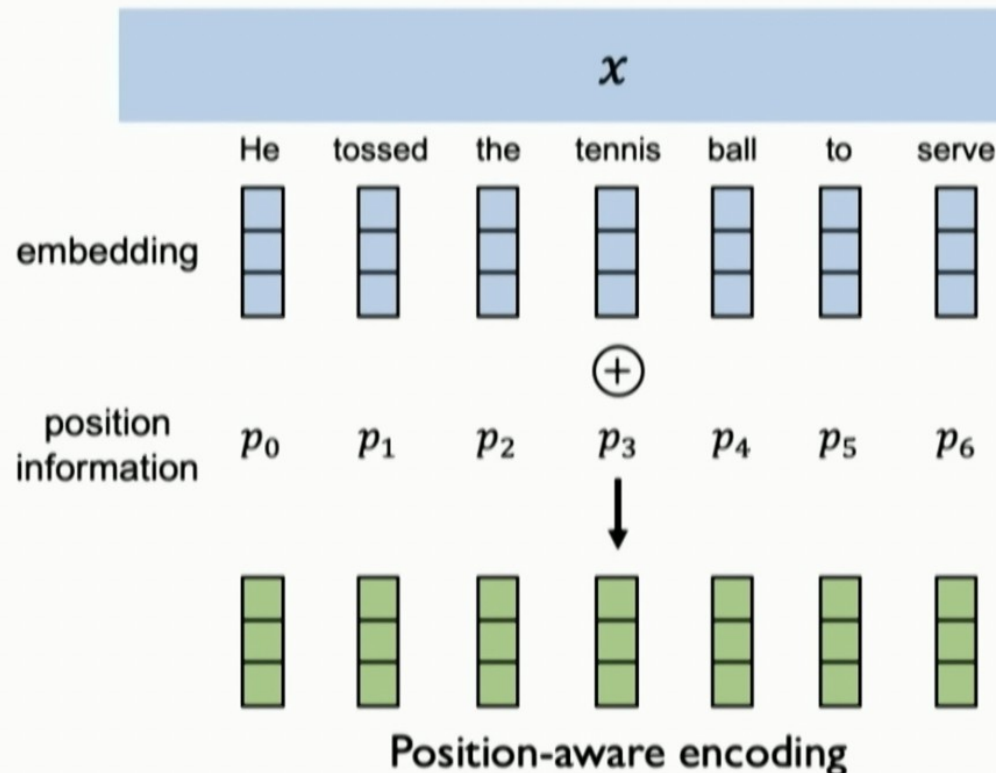
Transformers do not inherently process sequential data in order (like RNNs or LSTMs), they require a mechanism to understand the order or position of each element in the sequence.

This is where positional embeddings come in.

1. Encoding the **position**
2. Extracting **query, key, value**
3. Compute **attention weighting**
4. Extract **features with high attention**

This sum of **input embeddings + positional encodings/embeddings** is a complete representation of each input element:

It combines its semantic meaning (input embedding) ... with its position in the sequence (positional embedding)



Positional encoding (fixed sinusoidal vectors)

+

Input token embedding (learned)